WHITE PAPER · HWACHEON HI-ECO 35 (TWO CELLS) × AEROSPACE

Hwacheon Hi-Eco 35 for Aerospace

Mid-size CNC lathe pair for shafts, gear blanks, flanges, and threaded components with consistency and clean finishes. This paper covers what the machine does, the aerospace parts we produce with it, the alloys we run, the tolerances and finishes aerospace buyers expect, and the documentation package.

12.5" OD × 42" L max turning (each cell)
ENVELOPE

±0.0005" routine on critical features
TOLERANCE

1994
SINCE

Same-week / rig-down direct
RESPONSE



Ironwood Revival / Quickdraw Slim · 12.5" OD × 42" L max turning (each cell) · mid-size precision CNC turning center

PUBLISHED	LAST UPDATED	FORMAT
January 15, 2026	2026-01-15	Web + PDF

READ TIME

~6 min

Executive summary

Machine: Hwacheon Hi-Eco 35 (two cells) (shop nickname: *Ironwood Revival / Quickdraw Slim*). mid-size precision CNC turning center at B&R Productions, running from our New Waverly, Texas shop.

Application: Mid-size CNC lathe pair for shafts, gear blanks, flanges, and threaded components with consistency and clean finishes for Aerospace customers — Aerospace primes, tier-1 and tier-2 suppliers, MRO shops, propulsion component manufacturers, structural assembly integrators, actuator OEMs.

Envelope: 12.5" OD × 42" L max turning (each cell). **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

The Hi-Eco 35 pair is the mid-size backbone of the shop's turning capacity. Two identical cells means when one is loaded with a long-cycle Inconel job, the other keeps repair and small-batch work moving. Shafts, gear blanks, flanges, threaded components — the class of parts that make up the bulk of oilfield turning volume runs here.

Why Ironwood Revival / Quickdraw Slim for Aerospace work

Mid-size aerospace turned components in the 12-inch class — flanged fittings, housings, spacers, gear blanks — with two identical cells behind them. For aerospace work where a schedule slip cascades into someone else's build sequence, being able to move a proven program to the other cell without re-qualifying it is worth real money.

What this looks like on a real part

Inconel 718 in aerospace-aged condition is harder and less forgiving than the annealed billet stock most general job shops meet, and it is safe to assume aerospace 718 is aged unless the print says otherwise. Aged 718 punishes a dull edge immediately: the material work-hardens under a rubbing tool, and the next pass is cutting something harder than the one before it. Sharp tooling, committed feed, and flood coolant — and no drifting insert intervals across a production run.

When this is the wrong machine

42 inches between the extremes, and no live tooling. Longer aerospace shafting goes to Rattle & Hum; parts needing milled features referenced to a turned datum go to Torque & Tumble.

Full specifications

Model	Hwacheon Hi-Eco 35 — two cells: #16 ("Ironwood Revival") + #14 ("Quickdraw Slim")
Type	Mid-size CNC turning centers, run in parallel for throughput
Max turning capacity	12.5" OD × 42" L per cell (B&R working spec)
Platform max swing	23.6" over bed; 16.35" turning diameter platform-max
Platform max length	50" between centers
Chuck	12" 3-jaw hydraulic (15" available on some builds)
Turret	10-station
Control	Fanuc OT-C
Configuration	Two identical cells for parallel throughput
Tolerance capability	±0.0005" routine on critical features
Materials	Stainless, carbon steel, aluminum, Inconel, Duplex, 17-4 PH, Monel

Values marked "platform" reflect the machine manufacturer's published spec range for this platform family; B&R's working spec is what we quote on this specific unit.

Who this serves in Aerospace

Typical buyer: Aerospace primes, tier-1 and tier-2 suppliers, MRO shops, propulsion component manufacturers, structural assembly integrators, actuator OEMs.

What's hard about aerospace work: Certification-driven documentation, exotic superalloys, tight tolerances on structural and engine components, and finding a job shop that understands aged-condition superalloys rather than learning on the part.

Typical Aerospace parts we produce on Ironwood Revival / Quickdraw Slim

- Mid-size aerospace shafts, bushings, pins
- Actuator rods and cylinder components
- Engine and gearbox shafts
- Fittings and adapters (fluid and pneumatic)
- Landing gear turned components

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Aerospace

Standard range: Titanium Grade 5 (Ti-6Al-4V) and Grade 23 (Ti-6Al-4V ELI) · Inconel 718 · Inconel 625 · Waspaloy · A286 · 17-4 PH · 15-5 PH · 13-8 Mo PH · Aluminum 7075/6061 · Custom aerospace alloys on request.

Titanium Grade 5 is the workhorse and it's temperature-sensitive: aggressive chip load with flood coolant keeps the heat in the chip, not the workpiece. Light cuts overheat titanium and cook tools; that's why titanium demands a different intuition than steel. Inconel 718 in aerospace-aged condition is harder and less forgiving than the annealed billet stock — most aerospace 718 work assumes aged. 17-4 PH and 15-5 PH are common for structural and actuator components; both are precipitation-hardening stainless with condition-specific machining plans.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

How we machine it — our 5-step process

This is the process B&R runs on every job, aerospace or otherwise. The specifics of feeds, speeds, and tooling shift by alloy and machine; the discipline is universal.

1. **Material verification.** Every job starts with material traceability. Heat and lot numbers documented against the mill test report. For customer-supplied material, we verify against the CofC before touching the stock.

2. **First-article layout.** Stock is measured and laid out before any cutting starts. Concentricity, roundness, and squareness of the starting material established; if the stock is out of tolerance we call the customer before wasting time on a bad blank.

3. **Roughing with process control for the alloy.** Feeds, speeds, and coolant strategy set for the specific alloy family — sharp coated carbide (or PCD for high-volume Ti Grade 5), positive rake, high-pressure through-tool coolant on 2507 and Inconel to control cutting temperature. Interrupted cuts get extra attention because they cycle heat into the workpiece.

4. **Finish pass to spec.** Finish pass with a fresh insert, controlled DOC, and measured surface-finish targets. Critical features (bore concentricity, seat grooves, sealing surfaces) are held tighter than most job-shop lathes can offer.

5. **CMM verification + documentation.** Every critical dimension CMM-verified. First-article report generated. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, tell us with the RFQ and we'll be straight about what we can and can't produce in-house.

Tolerance and finish expectations in Aerospace

±0.0005" routine on critical features. Tighter with proper fixture and setup — some aerospace features run ±0.0002" or GD&T-driven cylindricity and position tolerances. CMM verification standard on first articles; process-driven measurement on runs. Surface finish measured, not estimated.

Documentation and quality workflow

First-article layout and CMM verification standard. Material traceability by heat and lot. Certificates of conformance issued with delivery. Proprietary prints handled under NDA — prints stay in-shop and aren't sent to outside vendors. To be clear about what we are not: B&R is not AS9100-registered and not ITAR-registered. If your program requires either, you need a registered shop and we'll say so on the first call rather than after you've placed the order.

Response time and emergency work

AOG (aircraft-on-ground) work prioritized when material is available. Emergency response supported for existing aerospace customers with blanket arrangements or established relationships. Call direct at (936) 291-7827.

How we compare

Compared to a captive aerospace shop: we're faster for one-offs and small runs, more flexible on setup, and we don't require a long approval process. Compared to a lowest-bidder commodity shop: we understand the alloys and we treat first-article inspection as a discipline rather than a formality. Where a registered shop beats us is on programs that require AS9100 flow-down — we're the wrong call for those, and the right call for non-controlled work, prototypes, tooling, and MRO parts where alloy competence and turnaround matter more than a certificate on the wall.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request, and prints stay in-shop rather than going out to third-party vendors. Note for defense and aerospace buyers: B&R is not ITAR- or AS9100-registered, so export-controlled prints and AS9100 flow-down programs need a registered shop — we'll tell you that on the first call, not after the PO.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Frequently asked questions

8 questions covering the Hwacheon Hi-Eco 35 platform and Aerospace work. Also indexed as FAQ schema for AI answer engines.

Why does B&R run two Hi-Eco 35 cells instead of one?

Parallel throughput. When one cell is loaded with a long-cycle Inconel job, the other keeps mid-size repair and small-batch work moving. Two identical cells means predictable capacity when volume matters.

What's the max turning capacity per cell?

12.5" OD × 42" L is B&R's working spec. The platform max is 16.35" turning diameter × 50" between centers with a 23.6" swing.

What chuck size and turret does the Hi-Eco 35 use?

12" 3-jaw hydraulic chuck (15" available on some builds). 10-station turret. Fanuc OT-C control.

Do you serve aerospace primes and tier-1/2 suppliers?

On work that doesn't require a registered QMS, yes — that's the honest boundary. B&R is not AS9100-registered, and where that matters it matters absolutely, so we say it before you spend time on an RFQ: if your program requires AS9100 flow-down, we're the wrong shop. Prototypes, tooling, fixtures, MRO and repair parts, and non-controlled components are where a shop like this is genuinely useful to a prime or a tier-1, and that's the work we'd like to quote.

Can you handle ITAR-controlled aerospace/defense work?

No. ITAR compliance requires registration with the U.S. State Department's DDTC, and B&R does not hold that registration. If your print is ITAR-controlled, you need a registered shop — send it elsewhere and we'd rather tell you now than after you've placed an order. Plenty of aerospace and defense work isn't export-controlled, and that work we quote gladly. Either way, proprietary prints are handled under NDA and stay in-shop.

What tolerances do you hold on aerospace parts?

±0.0005" routine on critical features. First-article layout and CMM verification standard. Surface-finish targets measured to spec.

What aerospace alloys do you run?

Titanium Grade 5 (Ti-6Al-4V), Grade 23 ELI, Inconel 718 and 625, Waspaloy, A286, 17-4 PH, 15-5 PH, aluminum 7075/6061. Custom aerospace alloys on request.

Can you respond to AOG (aircraft-on-ground) work?

Yes, when material is available. Emergency response supported for existing aerospace customers. Newer customers should call to discuss what's realistic for their specific alloy and part.

Related white papers

See all machine × industry white papers →

Ready to talk about your Aerospace work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

Published by B&R Productions — a precision CNC machining shop in New Waverly, Texas, in business since 1994. Serving oil & gas, aerospace, defense, and industrial customers across Texas and the Gulf Coast.

Written by the B&R Productions team. Machine specs verified against manufacturer data. Alloy and process notes reflect shop practice as of 2026-01-15.

Hwacheon Hi-Eco 35 (two cells) overview · All white papers · Aerospace industry page